

Piecewise Polynomial Regression - Data Analysis and Visualization

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Multiple Linear Regression (MLR)

- ▷ MLR can be used to describe the relationships between a set of explanatory or independent variables $(x_1, x_2, \dots, x_k)$ and a dependent variable (y)
- ▷ We are interested in the relationship between **each independent variable** and the dependent variable **after accounting for remaining independent variables**.
- ▷ MLR allows **quantifying the relationship** between our response variable and our explanatory variables as well as providing a tool for **predicting the response of a new observation** for a given set of values for $x_1, x_2, \dots$, and x_k .

Multiple Linear Regression (MLR)

The equation for the simple linear regression line is given by

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + e$$

- ▷ y is the response or dependent variable
- ▷ $x_1, x_2, \dots, x_k$ are the explanatory or independent variables
- ▷ β_0 is the intercept (the value of y when the $x_1, x_2, \dots, x_k$ are set to 0)
- ▷ β_1 is the slope (the expected change in y for each one-unit change in x_1 after adjusting for $x_2, \dots, x_k$)
- ▷ β_k is the slope (the expected change in y for each one-unit change in x_k after adjusting for $x_1, x_2, \dots, x_{k-1}$)
- ▷ e is the random error which we assume is normally distributed with a mean of 0 and a variance of σ^2

Regression Equation

The equation for the multiple linear regression line is given by

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k$$

where

- ▷ $\hat{y}$ (read y hat) is the expected or predicted value of y for a given values of $x_1, x_2, \dots, x_k$
- ▷ $\hat{\beta}_0$ is the least-squares estimates of (the intercept)
- ▷ $\hat{\beta}_1, \hat{\beta}_2, \dots$ and $\hat{\beta}_k$ are the least-squares estimates of $\beta_1, \beta_2, \dots$ and β_k respectively

Regression Equation

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$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k$$

In the least-squares regression, the estimates are selected in such a way that the following quantity is **minimized**:

$$(y - \hat{y})^2 = \left[y - (\hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2 + \dots + \hat{\beta}_k x_k) \right]^2$$

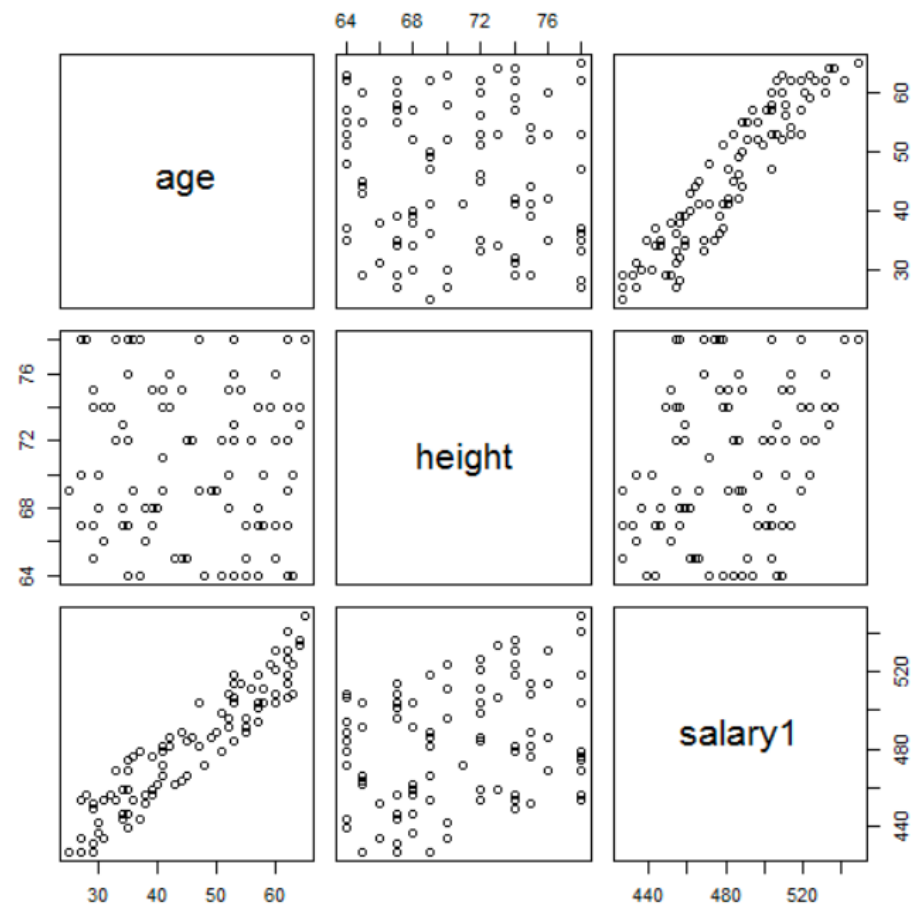
A MLR Example

In the book Blink by Malcolm Gladwell, Gladwell states that a study of CEOs of Fortune 500 companies found that these individuals tend to be taller than the average US population.

In order to study this phenomenon in more detail and to see if height is associated with increased success in business (as measured by salary), 100 men between the ages of 25 and 65 were polled for their heights (in inches) and annual salaries.

A MLR Example R Commands - Scatterplot Matrix

```
# Scatterplot Matrix  
> data <- read.csv("CEO_salary.csv")  
> attach(data)  
> salary1 <- salary/1000  
> data1 <- data.frame(age, height, salary1)  
> cor(data1)  
> pairs(data1)
```



Inference Global F test

- ▷ We use the ANOVA table and are interested in testing the **alternative hypothesis that at least one of the slope parameters is different than 0**.
- ▷ If this test confirms that there is at least one slope parameter that is different than 0, then **subsequent F-tests or t-tests** can be used to assess whether each individual slope parameter is different than 0.
- ▷ In MLR, the F-test for the model is referred to as the **global test**.

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Difference to SLR is that, here $k > 1$. The exact value of k depends on the number of variables in the model.

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F -statistic	p-value
Regression	Reg SS	Reg df = k	Reg MS = Reg SS/Reg df	$F = \text{Reg MS} / \text{Res MS}$	$P(F_{\text{Reg df, Res df}, \alpha} > F)$
Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS/Res df		
Total	Total SS = Reg SS + Res SS				

Inference - Anova Table Components

	SS (Sum of Squares)	df (degrees of freedom)	MS (Mean Square)	F-statistic	p-value
Regression	Reg SS	Reg df = k	Reg MS = Reg SS/Reg df	$F = \text{Reg MS}/\text{Res MS}$	$P(F_{\text{Reg df, Res df}, \alpha} > F)$
Residual	Res SS	Res df = $n - k - 1$	Res MS = Res SS/Res df		
Total	Total SS = Reg SS + Res SS				

- ▷ Reg SS = $\sum_{i=1}^n (\hat{y}_i - \bar{y})^2$, Res SS = $\sum_{i=1}^n (y_i - \hat{y}_i)^2$, Total SS = $\sum_{i=1}^n (y_i - \bar{y})^2$
- ▷ **Reg df = k** equals to the number of predictors in the model.
- ▷ **Res df = $n - k - 1$** equals to sample size minus the number of predictors in the model minus 1.
- ▷ Reg MS = Reg SS/Reg df (the regression mean square)
- ▷ Res MS = Res SS/Res df (the residual mean square)
- ▷ $F = \text{Reg MS}/\text{Res MS}$
- ▷ p-value = the probability that the observed value of test statistic or a more extreme value could have been observed by chance

Global F-Test

In MLR, the first formal tests of hypotheses is for the overall model.

They test the null hypothesis that

- ▷ all slope coefficients are equal to 0

$$(H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0)$$

The null hypothesis is the same as asserting that there is **no linear relationship between the response and explanatory variables.**

- ▷ alternative that at least one of the slope coefficients is different from zero
($H_1 : \beta_i \neq 0$ for at least one i).

The null hypothesis is rejected if there is at least one that is sufficiently far from 0 or (equivalently) a large majority of the total sum of squares is explained by the regression.

F-Test for MLR

$$F = \frac{\text{MS Reg}}{\text{MS Res}}$$

F-distribution with k and $n - k - 1$ degrees of freedom under H_0 .

The decision rule for a level α test is:

Reject $H_0 : \beta_1 = \beta_2 = \dots = \beta_k = 0$ if $F \geq F_{k, n-k-1, \alpha}$

Otherwise, do not reject H_0

where **$F_{k, n-k-1, \alpha}$ is the value from the F-distribution** with

- ▷ k degree of freedom (numerator) and
- ▷ $n - k - 1$ degrees of freedom (denominator) and
- ▷ associated with a right-hand tail probability of α .

MLR Inference t-test

If the overall model is significant, then the significance could be attributed to any one of the independent variables.

Perform testing on each individual parameter to identify the relative contribution of each independent variable.

In order to test each if $\beta_i = 0$ after controlling for the other independent variables in the model, we use a t statistic:

$$t = \frac{\hat{\beta}_i}{SE_{\hat{\beta}_i}}$$

where

$SE_{\hat{\beta}_i}$ the standard error of the estimate of (in the regression model with the other independent variables included) which follows a t-distribution with n-k-1 degrees of freedom under H_0 .

MLR Inference t-test

The decision rule for a two-sided level α test is:

- ▷ Reject $H_0 : \beta_i = 0$ if $|t| \geq t_{n-k-1, \alpha/2}$
- ▷ Otherwise, do not reject $H_0 : \beta_i = 0$

where

$$t_{n-k-1, \alpha/2}$$

is the value from the t-distribution with $n - k - 1$ degrees of freedom and associated with a right hand tail probability of $\alpha/2$.

MLR Inference t-test Confidence Interval

We can calculate the two-sided $100\%(1 - \alpha)$ confidence interval for using the following formula:

$$\hat{\beta}_i \pm t_{n-k-1, \frac{\alpha}{2}} \cdot SE_{\hat{\beta}_i}$$

Multi Linear Regression assumption

- Parameters are assumed no correlation
- How do we deal with interaction between parameters?
 - Example: increasing spending on radio advertisement increases effectiveness of TV advertisement?
 - Example: administering combined medicine for an illness.

Extending Linear Model

- Not assuming additive assumption
- Instead assuming interactions between parameters
- Example: Drug 1 and Drug 2 impact on managing blood pressure

$$y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

- Blood pressure as a function of Drug 1 (X_1) and Drug 2 (X_2)

Example of two Drugs

- Assume changing one drug changes of the other drug
- In other words, there is an interaction between parameters
- In this case we want to predict blood pressure after using both drugs

Capturing interactions

- We add a product term, which can be seen as adding one parameter as coefficient of the other parameter, like ...
 - In other words, this is the simplest interaction model, in which coefficient of one parameter is linear function of the other one.
- Interaction can be assessed separately with its own p-value and R squared.

Interaction between Qualitative & Quantitative Parameters

- Following the same logic, multiplication of parameters will be added, as follows
- Example:** predicting risk of heart attack based on blood pressure and race (simplifying to two class, e.g Latino vs not Latino)
 - X1 is blood pressure (quantitative)
 - X2 is race (qualitative)

$$R = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 (X_1 X_2)$$

$$R = \beta_0 + \beta_1 X_1 + \begin{cases} (\beta_2 + \beta_3 X_1) \\ 0 \end{cases}$$

Note: qualitative parameter would change the intercept of the multi linear regression line

$$R = \beta_0 + \beta_1 X_1 + \beta_2 X_2$$

or

$$R = \beta_0 + \beta_1 X_1 + \begin{cases} \beta_2 \\ 0 \end{cases}$$

While with interaction element not only the intersection, but also the slope of the line changes.

$$R = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_2 X_1$$

$$\text{Or } R = \beta_0 + (\beta_1 + \beta_3 X_2) X_1 + \beta_2 X_2$$

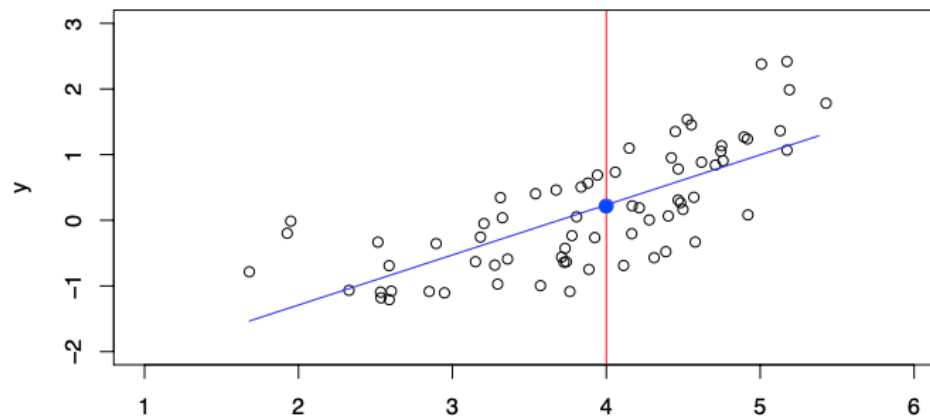
$$R = \beta_0 + \beta_1 X_1 + \begin{cases} \beta_2 + \beta_3 X_1 \\ 0 \end{cases}$$

Hierarchy Principle

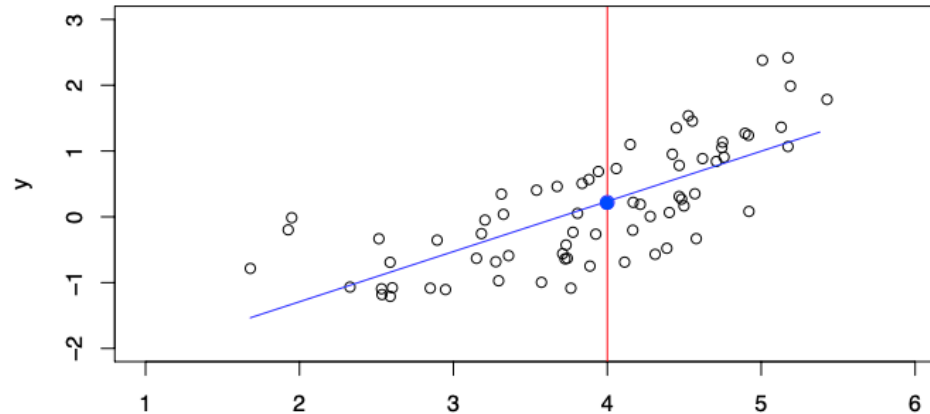
- What if interaction has a small p-value, but the main parameter don't?
- If we include the interaction, we have to include the main parameters as well – this is called “Hierarchy principle”
 - Intuition behind this: it will help to interpret the model. Otherwise, the interaction term will also capture the effect of the main parameter.

Polynomial Regression

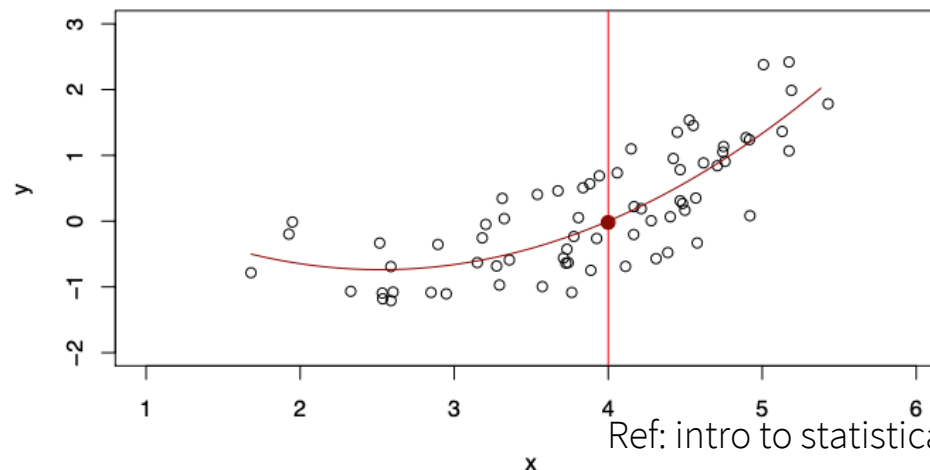
- A linear model $f(x) = \beta_0 + \beta_1 X$ is a good fit here



- A linear model $f(x) = \beta_0 + \beta_1 X$ is a good fit here



- A quadratic model $f(x) = \beta_0 + \beta_1 X + \beta_2 X^2$ fits even better



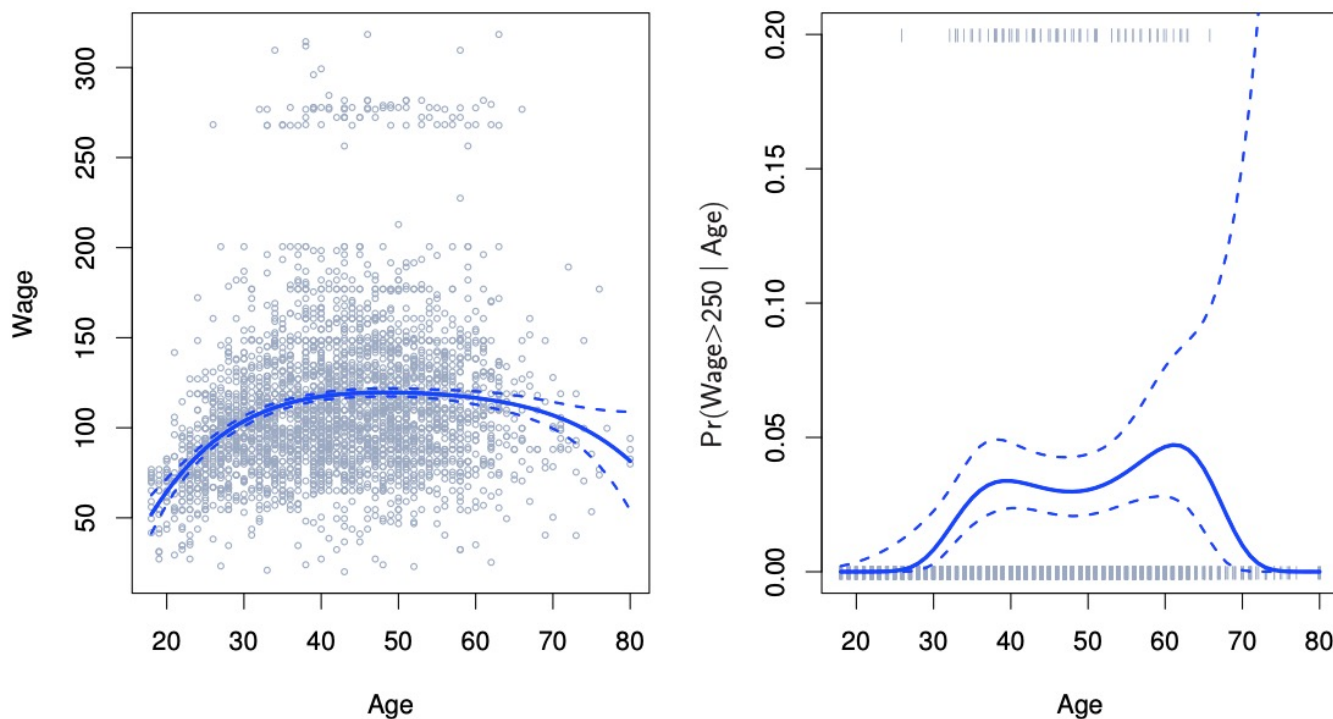
Why Non-Linear Model?

- Real data is never linear, but linear is a good approximation
- If not, polynomial regression is a good extension
 - Flexible
 - Easy
 - Interpretable

Example of Polynomial Regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \beta_3 x_i^3 + \dots + \beta_d x_i^d + \epsilon_i$$

Degree-4 Polynomial



Ref: intro to statistical learning, G. James, et. al.

Polynomial Regression

- Same as multi-linear regression, except higher degree of parameters also get included.

$$f(x) = \beta_0 + \beta_1 X + \cdots + \beta_n X^n$$

- Since $f(x)$ is a linear combination of β values, we can compute pointwise variance of the function – $\text{Var}[f(x_0)]$ and also find confidence interval for prediction at point x_0

What Degree is good?

- What degree should be used? We mostly don't go above the power of 3.

Draw backs of Polynomial Regression

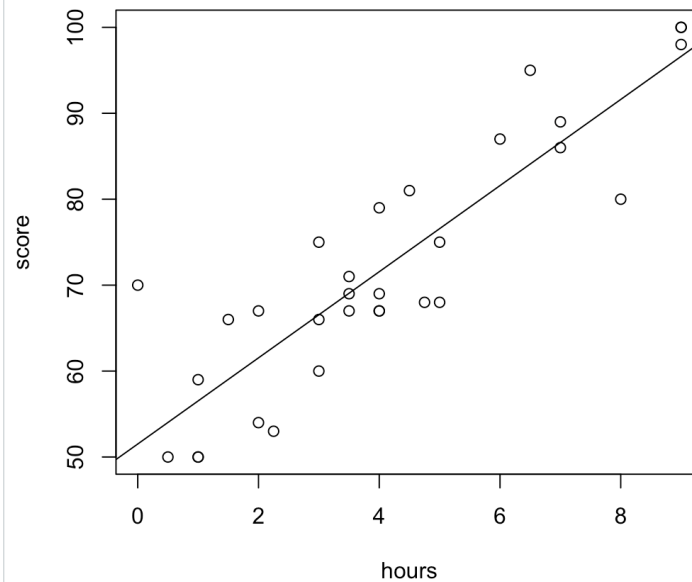
- Overfitting
- Polynomial regression is very bad for extrapolation
 - Note that Polynomial regression might behave widely at the end of the range, since mostly the data is thin at the end and the model tries to match the data in other places.

Example of Overfitting

Data: number of hours studied and score of 31 students.

Linear regression estimate

	name	hours	score
1	Allen	8.0	80
2	Brown	2.0	67
3	Cole	9.0	98
4	Collins	9.0	100
5	Cooper	7.0	86
6	Cox	6.5	95



Learning From the Entire Data

- Perfect training with zero error

$R^2 = 1!$ PERFECT!

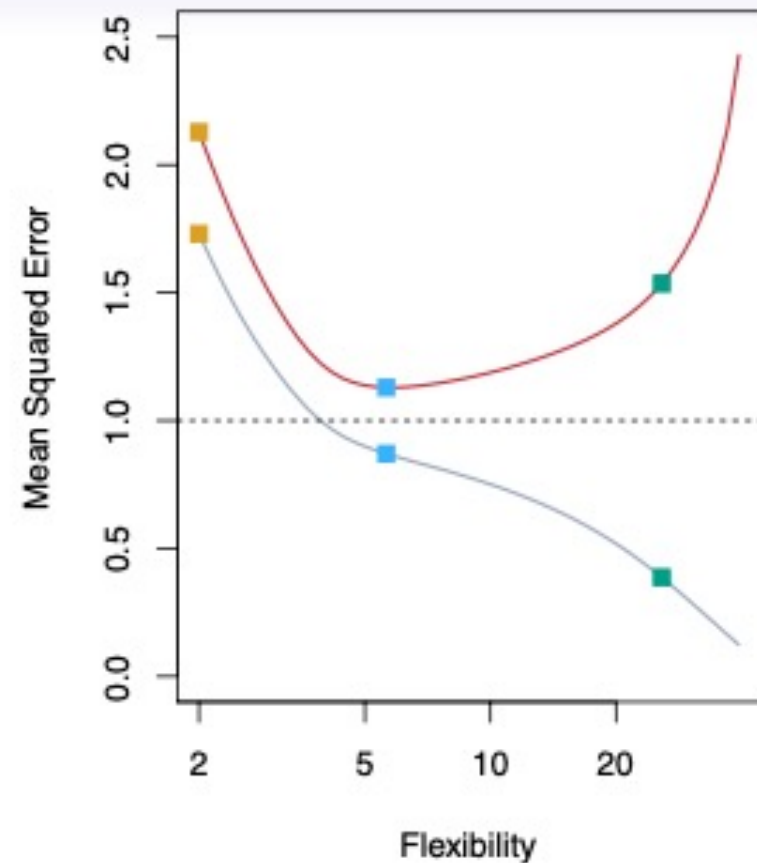
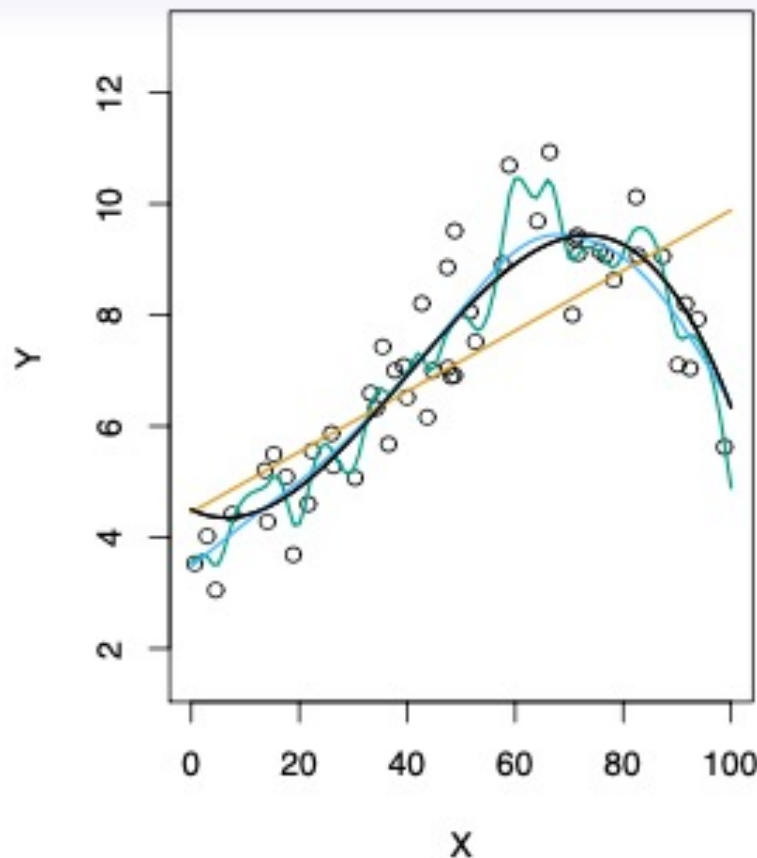
```
Residuals:
ALL 31 residuals are 0: no residual degrees of freedom!

Coefficients: (2 not defined because of singularities)
              Estimate Std. Error t value Pr(>|t|)
(Intercept)      80             NA      NA      NA
nameBrown       -13             NA      NA      NA
nameCole         18             NA      NA      NA
nameCollins      20             NA      NA      NA
nameCooper        6             NA      NA      NA
nameCox          15             NA      NA      NA
nameHall        -13             NA      NA      NA
nameHans        -21             NA      NA      NA
nameHoward        9             NA      NA      NA
nameJeffers     -10             NA      NA      NA
nameJohnson    -30             NA      NA      NA
nameJones       -14             NA      NA      NA
nameKing         1             NA      NA      NA
nameKnight      -11             NA      NA      NA
nameLee          -9             NA      NA      NA
nameMartin      -14             NA      NA      NA
nameMiller       -5             NA      NA      NA
nameMoore       -20             NA      NA      NA
nameMorris      -13             NA      NA      NA
nameMurphy       7             NA      NA      NA
nameReed        -5             NA      NA      NA
nameSmith       -30             NA      NA      NA
nameStewart     -12             NA      NA      NA
nameTaylor      -27             NA      NA      NA
nameThomas      -26             NA      NA      NA
nameThompson    -11             NA      NA      NA
nameWalker      -13             NA      NA      NA
nameWard         20             NA      NA      NA
nameWilliams    -30             NA      NA      NA
nameWright      -12             NA      NA      NA
nameYoung       -1             NA      NA      NA
hours           NA             NA      NA      NA
id              NA             NA      NA      NA

Residual standard error: NaN on 0 degrees of freedom
Multiple R-squared: 1, Adjusted R-squared: NaN
```

Overfitting - Training vs Test Data

- The approach minimizes MSE on a training set, e.g. with polynomial regression finds a curve that minimizes MSE.
- But the goal is having a model for the entire world!
 - Or how well the model works on a new data - “**Test Data**”.
- We know - the smallest training MSE does not guarantee the smallest test data MSE!



Reference: An Intro to Statistical Learning, G. James

- Black curve – left: actual model
- Gray curve – right: training error
- Red curve – right: testing error
- Blue square: Linear & Green square: Polynomial Reg

Trade-offs in the models

- Prediction accuracy versus interpretability.
 - Linear models are easy to interpret; more complex models are not.
- Good fit versus over-fit or even under-fit.
 - How do we know when the fit is just right?
- Parsimony versus black-box.
 - We often prefer a simpler model involving fewer variables over a black-box predictor involving all of them.

Bias-Variance Trade-off

- The examples of test versus training MSE's shows an important tradeoff of choosing a model.

Bias vs Variance

- Variance is variability of the model over different training set
- Bias is the expected difference between the model and actual value or $E[\hat{f}(x_0)] - f(x_0)$
- These are two competing choices

Bias of Learning Methods

- Bias definition - the expected difference between the model and the actual value

$$E[\hat{f}(x_0)] - f(x_0)$$

- E.g. linear regression assumes linear relationship between Y and X, which is mostly not correct.

The more complex a method, the more flexible and less bias, generally.

Model Variance

- Variance is variability of the model over different training set

The more complex a method, the more variance it has, generally.

- Suppose a model $\hat{f}(x)$ is fit to a training data, and let (x_0, y_0) be a test observation point.
- If the true model is $y = f(x) + \varepsilon$ with

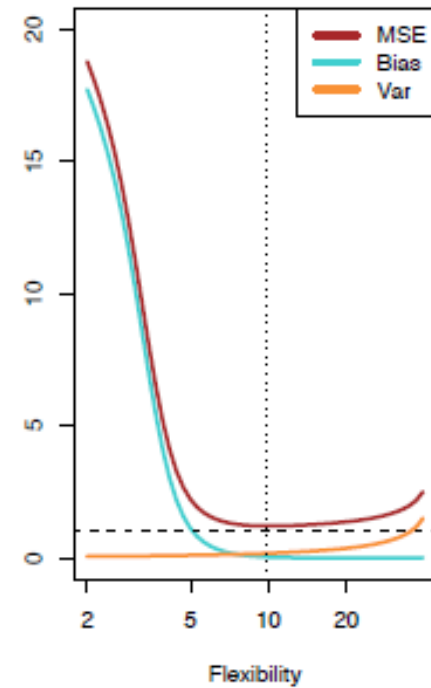
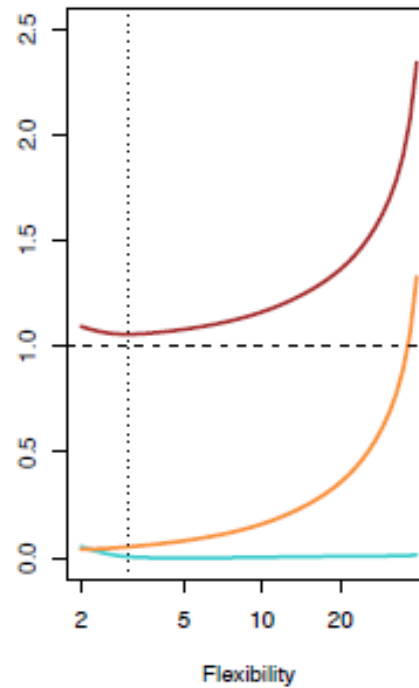
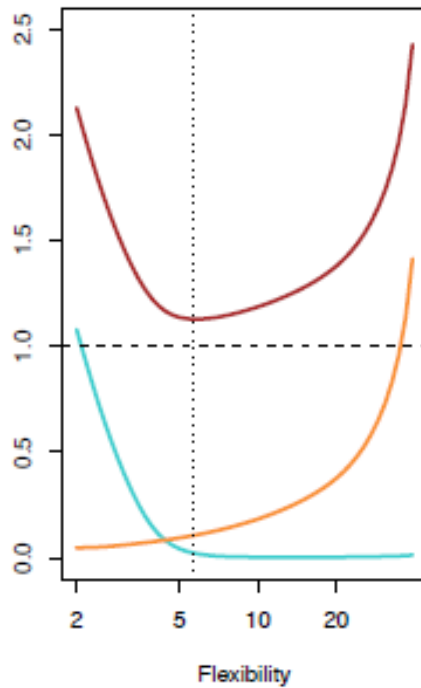
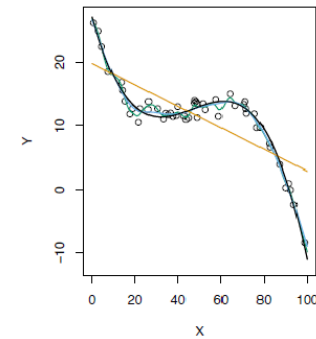
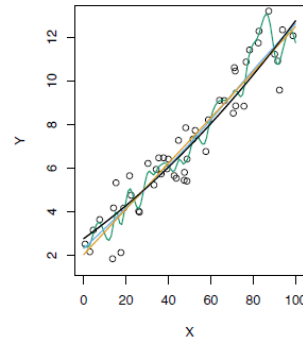
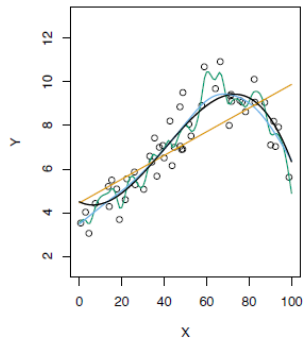
$$f(x) = E[Y|X = x]$$
- Then the expected MSE is

$$E[(y_0 - \hat{f}(x_0))^2] = \text{bias}^2 + \text{var}[\hat{f}(x_0)] + \text{var}(\text{noise})$$
- Where

$$\text{bias} = E[\hat{f}(x_0) - f(x_0)] = E[\hat{f}(x_0)] - f(x_0)$$

$$\text{var}[\hat{f}(x_0)] = E[\hat{f}(x_0) - y_0]^2$$
- Typically as the flexibility of the model increases, its variance increases, and its bias decreases.

Test MSE, Bias & Variance



Ref: In-depth introduction to machine learning, by T. Hastie and R. Tibshirani

Logistic function with Polynomial

$$\frac{\exp(\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d)}{1 + \exp(\beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d)}.$$

Step Functions

- “Using polynomial functions of the features as predictors in a linear model imposes a global structure on the non-linear function of X . We can instead use step functions in order to avoid imposing such a global structure.”
- Using step functions, create cut points ($C_1, C_2, \dots, C_K$), in the range of X and construct $K+1$ new variables:

$$\begin{aligned}C_0(X) &= I(X < c_1), \\C_1(X) &= I(c_1 \leq X < c_2), \\C_2(X) &= I(c_2 \leq X < c_3), \\&\vdots \\C_{K-1}(X) &= I(c_{K-1} \leq X < c_K), \\C_K(X) &= I(c_K \leq X),\end{aligned}$$

- $I(.)$ is an indicator function that returns a 1 if the condition is true.

Step Functions

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \dots + \beta_K C_K(x_i) + \epsilon_i.$$

$$\frac{\exp(\beta_0 + \beta_1 C_1(x_i) + \dots + \beta_K C_K(x_i))}{1 + \exp(\beta_0 + \beta_1 C_1(x_i) + \dots + \beta_K C_K(x_i))}$$

Piecewise Polynomial

- What if we know there is an underlying difference between X and Y relationship in different region of domain?
- For example,
 - Below 18 years old and above 18 years old react differently to a drug
 - Temperature behave differently in urban area and non-urban area

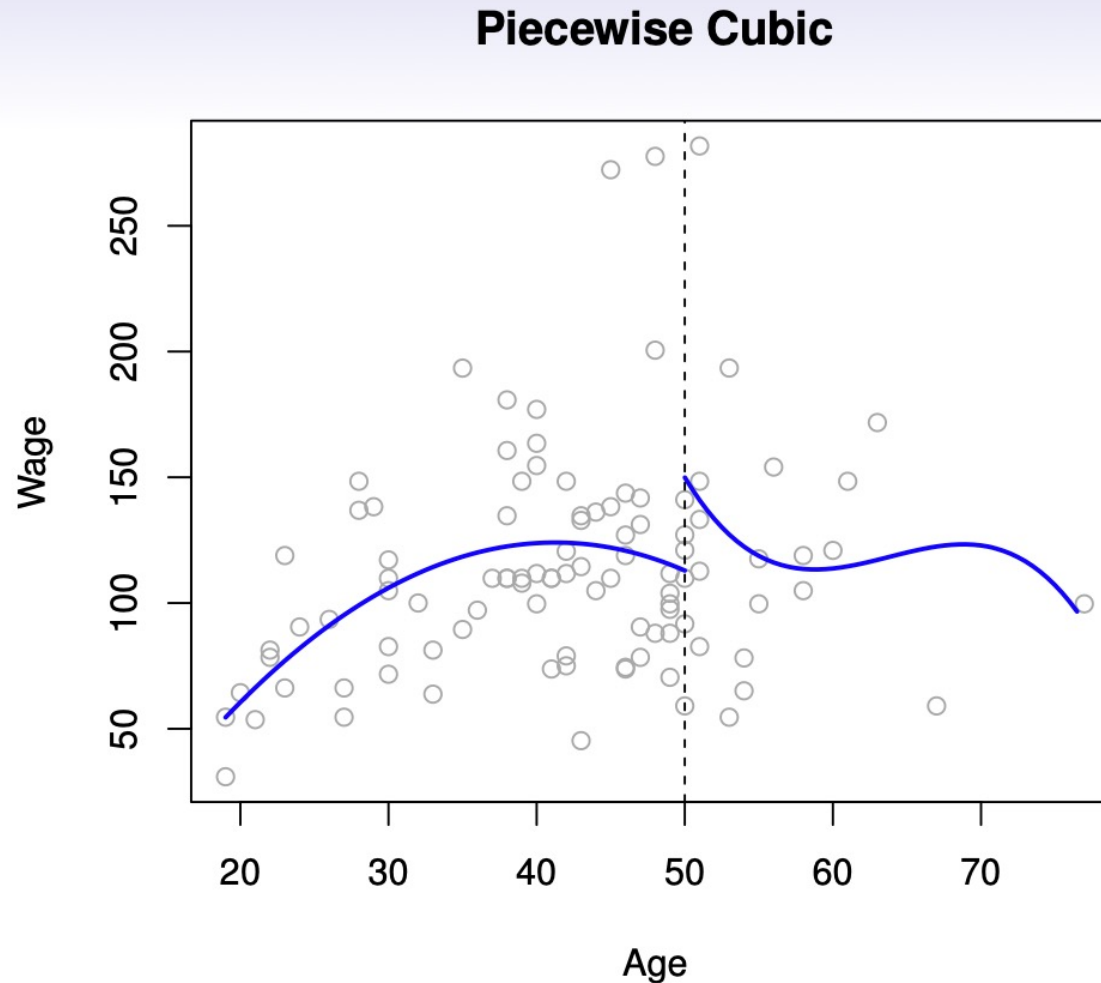
Piecewise Polynomial

- Dividing the domain of X to multiple regions and fit different polynomial to each section.
- For example, dividing the domain at point C and fit two quadratic functions

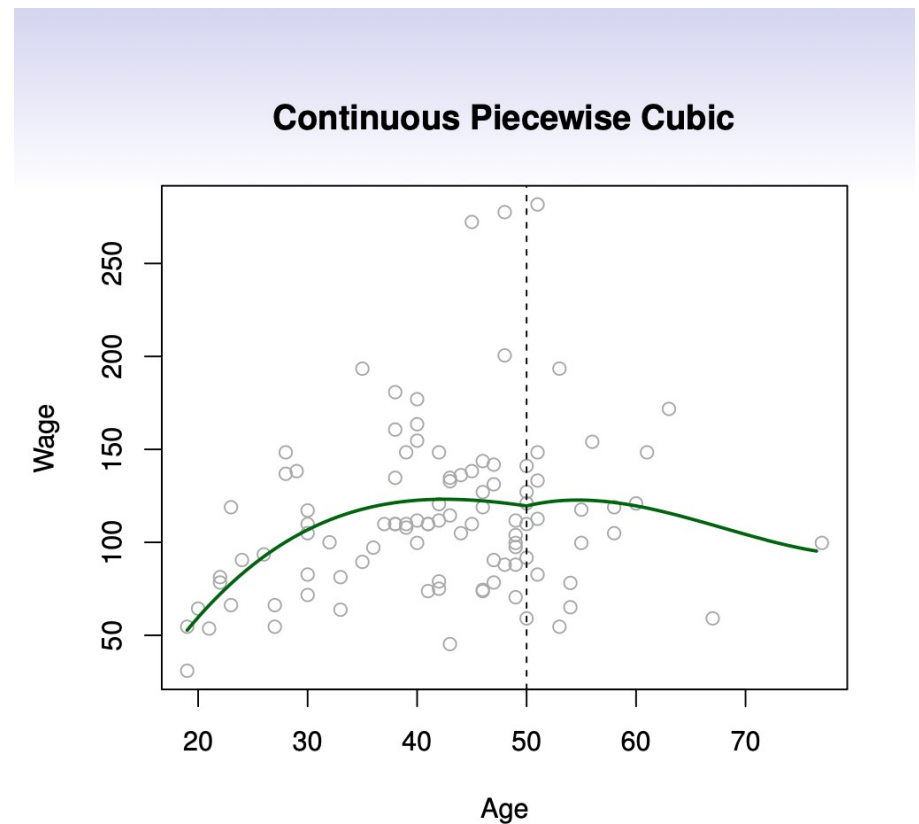
$$y = \begin{cases} \beta_{01} + \beta_{11}X + \beta_{21}X^2, & x < C \\ \beta_{02} + \beta_{12}X + \beta_{22}X^2, & x \geq C \end{cases}$$

Note: C is a constant which gets selected naturally based on the nature of the data

Piecewise Polynomial Example

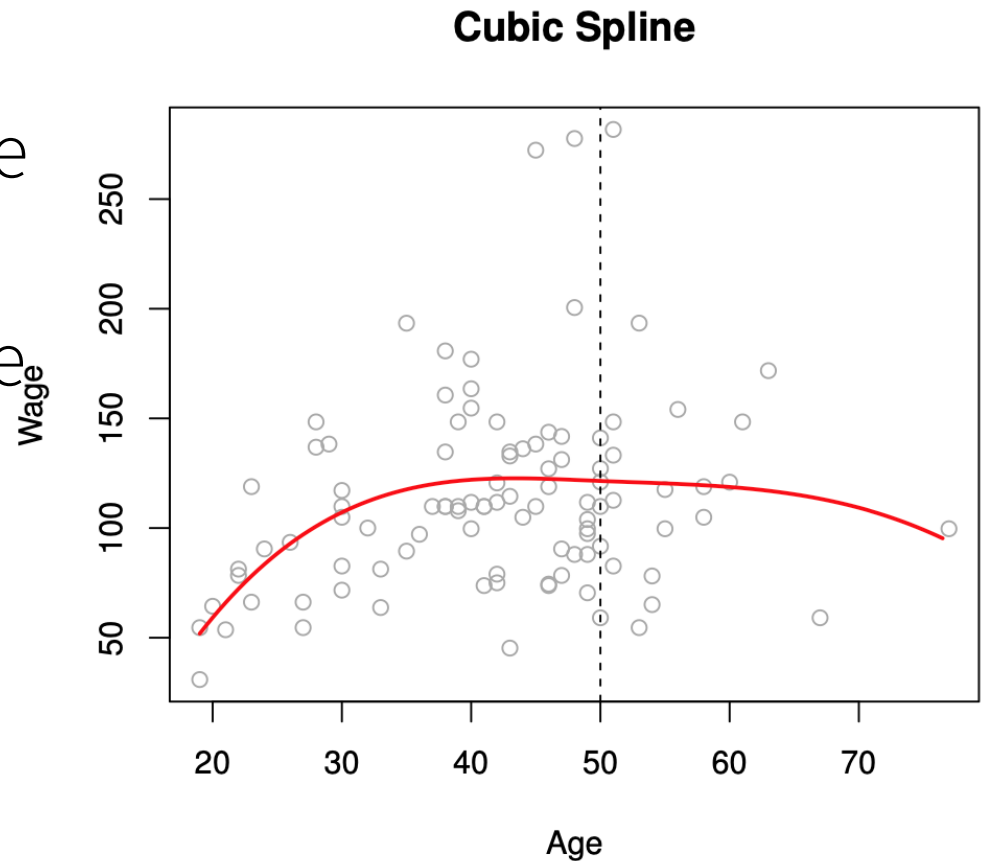


- Piecewise with forcing the same value at the junctions
- The curve won't be smooth



Spline

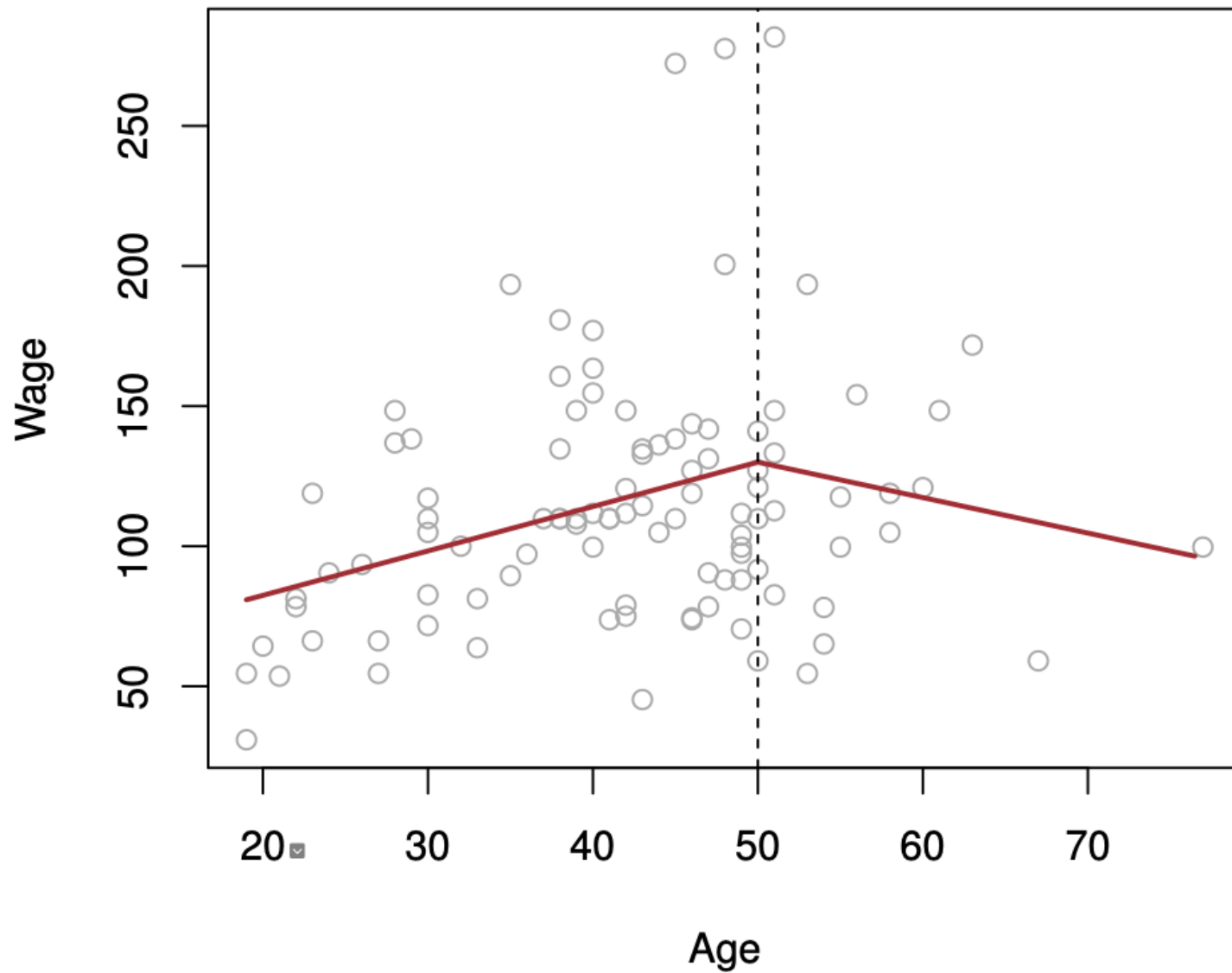
- The value are the same at the junctions
- Also derivatives are the same at the junctions



Splines

- For the order K polynomial Spline, up to the $(k-1)$ derivative can be added as constraints to be equal. Otherwise, the curves will become the same
- For example: for cubical Spline (degree of 3 polynomial), only up to the second derivative can be added as a constraint to be equal

Linear Spline



Ref: In-depth introduction to machine learning, by T. Hastie and R. Tibshirani

Natural Spline

- Natural Spline is a version of Spline, in which outside the range of the data gets extrapolated linearly
 - This translates to adding to constraint to each end of the range of the data, which are
 1. Continuity of the line to the last polynomial
 2. Equal first derivative at the junction

Smoothed Splines

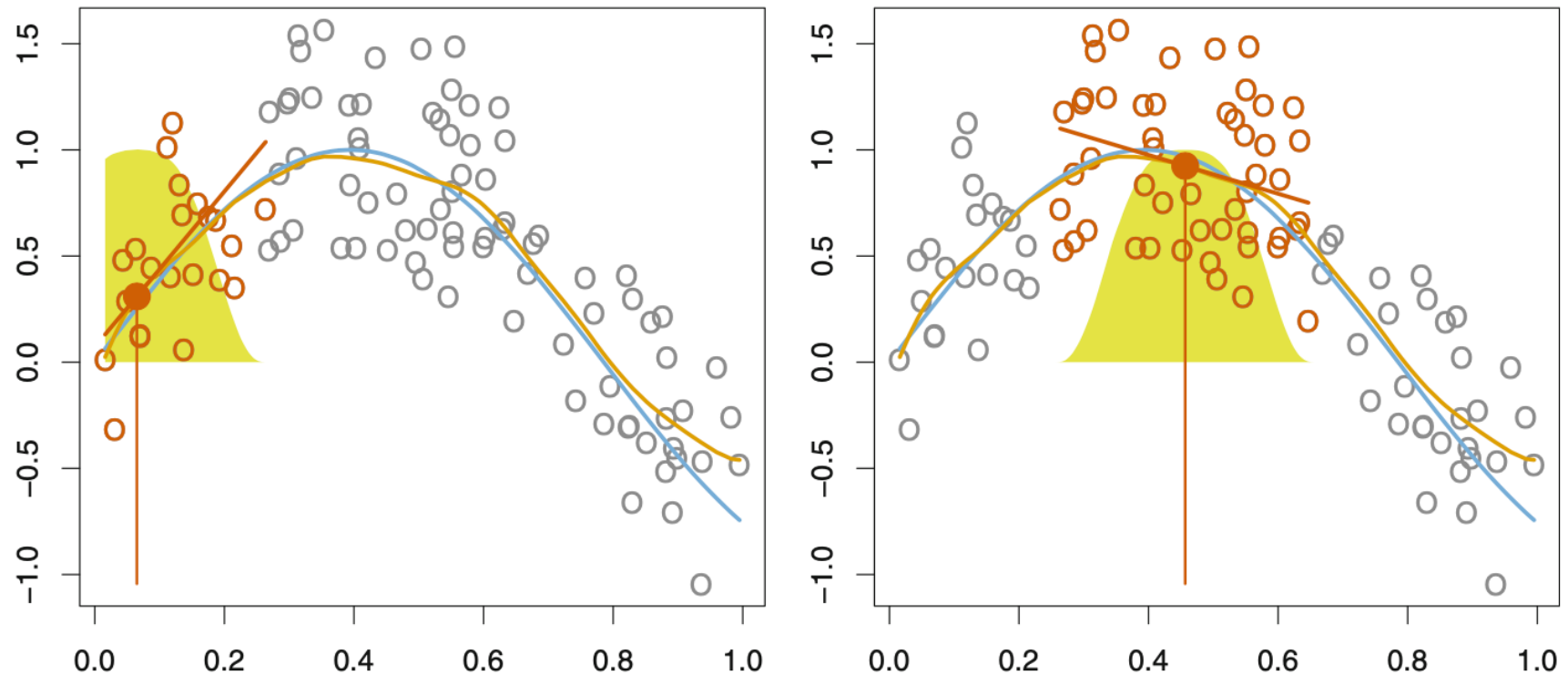
- Expanding Spline idea
- Integrating Smoothing function into the criteria as follows

$$\min_{g \in \mathcal{S}} \sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(m)^2 dm$$

- The first term is the same as least square error
- The second term is reducing roughness of the curve and reduces variation of the model
- Parameter λ is the only hyperparameter
- Note as $(\lambda \rightarrow \infty)$ the optimum model is a line, and if $(\lambda = 0)$ the model is normal polynomial regression
- The smoothed Spline is like adding knot at every unique sample

Local Regression

Local Regression



Local Regression Algorithm

1. Gather the fraction $s = k/n$ of training points whose x_i are closest to x_0 .
2. Assign a weight $K_{i0} = K(x_i, x_0)$ to each point in this neighborhood, so that the point furthest from x_0 has weight zero, and the closest has the highest weight. All but these k nearest neighbors get weight zero.
3. Fit a *weighted least squares regression* of the y_i on the x_i using the aforementioned weights, by finding $\hat{\beta}_0$ and $\hat{\beta}_1$ that minimize

$$\sum_{i=1}^n K_{i0} (y_i - \beta_0 - \beta_1 x_i)^2. \quad (7.14)$$

4. The fitted value at x_0 is given by $\hat{f}(x_0) = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Spline in R

- R has a package “Spline”, which has a function for Spline, called “`bs(x,..)`” and also a function for natural Spline, called “`ns(x,...)`”
- Smoothing Spline function is “`smooth.spline(x,y,lambda)`”